

Color Sensor

Objective:

To read and analyze RGB (Red, Green, Blue) color values using the built-in APDS9960 color sensor on the Arduino Nano 33 BLE Sense.

(Note: install `Arduino_APDS9960` Library)

Procedure:

1. Opened Arduino IDE.
2. Installed the `Arduino_APDS9960` library.
3. Opened the `ColorSensor` example from:
4. File → Examples → `Arduino_APDS9960` → `ColorSensor`
5. Uploaded the program to the Arduino Nano 33 BLE board.
6. Opened the Serial Monitor at 9600 baud rate.
7. Placed different colored objects near the sensor.
8. Observed the RGB values displayed in the Serial Monitor

Working Principle:

The APDS9960 color sensor detects the intensity of Red, Green, and Blue light reflected from an object.

The sensor measures:

- R → Red intensity
- G → Green intensity
- B → Blue intensity

The combination of these three values determines the detected color

Observations:

- RGB values changed when different colored objects were placed near the sensor.
- A red object produced a higher Red value.
- A green object produced a higher Green value.
- A blue object produced a higher Blue value.
- The sensor response depended on ambient lighting conditions.

$$r = 3$$

$$g = 2$$

$$b = 3$$

$$r = 20$$

$$g = 20$$

$$b = 25$$